

# Mahathir Mohammad Shuvo

Full-Stack Software Engineer

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## SUMMARY

Full-stack engineer with six years of professional experience shipping production systems across web, mobile and desktop. Works daily in TypeScript and Python across four concurrent production codebases - REST and WebSocket backends, React and React Native front ends and an Electron desktop app - and also writes Java and C++. Owns systems end to end: schema and API design, real-time features, payments, CI to the app stores, automated testing and production monitoring. Open-source contributor with seven pull requests merged upstream into PDFKit and Hono.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, Java, C++  
**Backend:** NestJS, Node.js, Express, Flask, Django, Prisma, REST APIs, WebSockets, Socket.IO, Zod  
**Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, SQLAlchemy  
**Frontend:** React, TanStack Router/Query/Table/Form, Redux, Zustand, Tailwind CSS, Radix UI, Vite  
**Mobile / Desktop:** React Native (Expo), Electron, Codemagic CI, EAS Build  
**AI:** Model Context Protocol (MCP), LangGraph, Anthropic API, RAG, Hugging Face Transformers, TensorFlow.js  
**Cloud / DevOps:** AWS, Google Cloud Platform, Docker, nginx, Apache, PM2, GitHub Actions, Vercel, Netlify  
**Security / Testing:** JWT, OAuth, bcrypt, WebCrypto; Playwright, Vitest, Jest, Mocha  
**Other:** WebRTC (mediasoup, LiveKit), web scraping, Google Maps API, Stripe, PDF and Excel generation

## EXPERIENCE

### Software Engineer

Apr 2024 - Present

Confidential client under NDA - Remote

- Develop a large TypeScript monorepo spanning HTTP backend, web app, admin console, mobile and desktop clients, with roughly a quarter of the codebase being automated tests.
- Ship a cross-platform Electron desktop app with signed auto-updates and PDF processing; integrate biometric identity verification (AWS Rekognition, Amplify Liveness, TensorFlow.js).
- Diagnose and fix defects across unfamiliar third-party code, and implement metered billing, subscriptions and payouts with Stripe against a rate-limited public API with key management.

### Full-Stack Developer

2020 - Mar 2024

DKERPS - Remote

- Sole backend developer across five products, owning deployment, testing and architecture on several concurrently.
- Built a Flask/SQLAlchemy large-file transfer service supporting single transfers up to 40 GB, on GCP with Ubuntu and Apache.
- Built Dropbox-style cloud storage with shared folders, business accounts and per-member permissions, on AWS with nginx at 99% uptime; and a centralised log-management system on Elasticsearch.
- Built a Zoom-style conferencing app (MERN, mediasoup WebRTC) and delivered a React Native (Expo) scheduling app - 100% of backend, 70% of frontend.

## PROJECTS

- Discount Bajar (discount-bajar.com) - live e-commerce platform. Four apps on one NestJS/Prisma/PostgreSQL backend: React web, admin console, Expo app for iOS and Android, and workers. Double-entry accounting with bank reconciliation, rider dispatch, referral and reward engine, recommendation engine, RAG product search, real-user monitoring.
- AI Employees - self-hosted multi-agent platform. Agents organised as a company collaborating over the Model Context Protocol; LangGraph orchestration with a PostgreSQL checkpointer, Redis-backed Socket.IO, React UI with Monaco editor and node-graph monitoring.

## OPEN SOURCE

- PDFKit (foliojs/pdfkit, 10.7k stars) - six merged pull requests: AcroForm date fields emitting invalid JavaScript so formatting never applied; indentAllLines applying the indent repeatedly across paragraphs and page breaks; table cell fonts inheriting an unrelated family; undefined option values serialising as the literal token; a hole in a sparse array shifting every later entry down one; and a missing embedded-file ref writing an unparseable token into the name tree.
- Hono (honojs/hono, 31.8k stars) - one merged pull request: an undefined entry inside a query array going out on the wire as the literal string undefined.
- Radix UI (19.2k stars) - open pull request fixing DismissableLayer requiring two outside clicks to close after inner content stopped event propagation.

**Languages:** Bengali (native), English (professional), Hindi (conversational)